# FASTIFY CHEAT SHEET

Core setup, initialization, operations & integration quick reference

PART 1: SETUP & CORE

## 01. SETUP & INSTANCE

Installation

```
npm install fastify
import Fastify from 'fastify'
const fastify = Fastify({ logger: true })
await fastify.listen({ port: 3000, host: '0.0...
fastify.log.info('Server started')
```

Fastest Node.js HTTP framework (benchmarks #1 2024)

## 02. ROUTES

Architecture

```
fastify.get('/users', async (request, reply) ...
return { users: [] }  // auto-serialized
})
fastify.post('/users', {
schema: { body: userSchema, response: { 201: ...
}, async (req, reply) => {
reply.status(201).send(newUser)
})
```

## 03. SCHEMA VALIDATION

YAML / Config

```
const schema = {
body: {
type: 'object',
required: ['name','email'],
properties: {
name: { type: 'string', minLength: 2 },
email: { type: 'string', format: 'email' }
}
}
}
```

JSON Schema via Ajv; also boosts serialization 2×

## 04. HOOKS

Workflow

```
fastify.addHook('onRequest', async (req, repl...
// auth check
})
fastify.addHook('preHandler', async (req, rep...
// rate limiting
})
fastify.addHook('onSend', async (req, reply, ...
```

Hooks: onRequestpreParsingpreValidationpreHandleronSendonResponse

## 05. PLUGINS

CRUD API

```
import fp from 'fastify-plugin'
const myPlugin = fp(async (fastify, opts) => {
fastify.decorate('db', new DB(opts))
})
fastify.register(myPlugin, { uri: process.env...
fastify.register(require('@fastify/cors'))
fastify.register(require('@fastify/jwt'), { s...
```

fp() makes plugin available across encapsulation boundaries

## 06. DECORATORS

Integration

```
fastify.decorate('config', { port: 3000 })
fastify.decorateRequest('user', null)
fastify.decorateReply('sendError', function(c...
this.status(code).send({ error: msg })
})
// In handler:
request.user  /  reply.sendError(401, 'Unauth...
```

# SECURITY, MONITORING & OPERATIONS

Production hardening, observability, performance tuning & CLI reference

PART 2: OPS & SECURITY

## 07. REQUEST & REPLY

Hardening

```
request.body / request.params / request.query
request.headers / request.ip / request.id
reply.send(data)  // infers Content-Type
reply.status(404).send({ error: 'Not found' })
reply.type('text/html').send('<h1>Hi</h1>')
reply.redirect('/new-url')
reply.header('X-Custom', 'value')
```

## 08. ERROR HANDLING

Observability

```
fastify.setErrorHandler(async (error, request...
request.log.error(error)
reply.status(error.statusCode ?? 500).send({
error: error.message
})
})
throw fastify.httpErrors.notFound('User not f...
// Requires @fastify/sensible plugin
```

## 09. LOGGING

Optimization

```
const fastify = Fastify({
logger: {
level: 'info',
transport: { target: 'pino-pretty' }
}
})
request.log.info({ userId: id }, 'Fetching us...
fastify.log.error({ err }, 'Fatal error')
```

Built on Pino  very fast structured logging

## 10. COMMON PLUGINS

Unit Tests

```
@fastify/jwt     JWT auth
@fastify/cors    CORS headers
@fastify/static  static files
@fastify/swagger  OpenAPI docs
@fastify/rate-limit  rate limiting
@fastify/compress  gzip/deflate
@fastify/sensible  httpErrors helpers
```

## 11. COMMON GOTCHAS

Commands

Schema must use additionalProperties:false or extra props pass
Decorated properties must be initialized in decorateRequest/Reply
Plugin encapsulation: fp() needed to share across contexts
reply.send() after async handler = duplicate response error
JSON schema 'format' requires ajv-formats plugin